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Clg Name :- T.B.Girwalkr Polytechnic , Ambajogai

Domain :- AI - ML

Branch Name :- Computer Engineering

1) Load the Ford Car Dataset using pandas.
   Display the first 10 rows and last 5 rows.
   Check the shape (number of rows and columns) and data types of all columns.
   Write observations about the dataset structure in comments.

```
df.head(10)
```

	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize
0	Fiesta	2017	12000	Automatic	15944	Petrol	150	57.7	1.0
1	Focus	2018	14000	Manual	9083	Petrol	150	57.7	1.0
2	Focus	2017	13000	Manual	12456	Petrol	150	57.7	1.0
3	Fiesta	2019	17500	Manual	10460	Petrol	145	40.3	1.5
4	Fiesta	2019	16500	Automatic	1482	Petrol	145	48.7	1.0
5	Fiesta	2015	10500	Manual	35432	Petrol	145	47.9	1.6
6	Puma	2019	22500	Manual	2029	Petrol	145	50.4	1.0
7	Fiesta	2017	9000	Manual	13054	Petrol	145	54.3	1.2
8	Kuga	2019	25500	Automatic	6894	Diesel	145	42.2	2.0
9	Focus	2018	10000	Manual	48141	Petrol	145	61.4	1.0

```
df.tail()
```

	model	year	price	transmission	mileage	fuelType	tax	mpg	engineSize
17961	B-MAX	2017	8999	Manual	16700	Petrol	150	47.1	1.4
17962	B-MAX	2014	7499	Manual	40700	Petrol	30	57.7	1.0
17963	Focus	2015	9999	Manual	7010	Diesel	20	67.3	1.6
17964	KA	2018	8299	Manual	5007	Petrol	145	57.7	1.2
17965	Focus	2015	8299	Manual	5007	Petrol	22	57.7	1.0

```
df.shape
```

```
(17966, 9)
```

```
df.dtypes
```

	0
model	object
year	int64
price	int64
transmission	object
mileage	int64
fuelType	object
tax	int64
mpg	float64
engineSize	float64

dtype: object

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17966 entries, 0 to 17965
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   model         17966 non-null  object 
 1   year          17966 non-null  int64  
 2   price         17966 non-null  int64  
 3   transmission  17966 non-null  object 
 4   mileage       17966 non-null  int64  
 5   fuelType      17966 non-null  object 
 6   tax           17966 non-null  int64  
 7   mpg           17966 non-null  float64
 8   engineSize    17966 non-null  float64
dtypes: float64(2), int64(4), object(3)
memory usage: 1.2+ MB
```

2) Perform a detailed check on the dataset:

Find the total number of missing values in each column.

Identify and remove duplicate rows if any.

Write a summary of your findings and how you handled them in comments.

```
df.isnull().sum()
```

	0
model	0
year	0
price	0
transmission	0
mileage	0
fuelType	0
tax	0
mpg	0
engineSize	0

dtype: int64

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.duplicated().sum()
```

```
np.int64(154)
```

3) Generate the statistical summary of all numeric columns using describe(). Identify minimum, maximum, mean, and median values for key columns like price, mileage, and year.

```
df.describe()
```

	year	price	mileage	tax	mpg	engineSize
count	17966.000000	17966.000000	17966.000000	17966.000000	17966.000000	17966.000000
mean	2016.866470	12279.534844	23362.608761	113.329456	57.906980	1.350807
std	2.050336	4741.343657	19472.054349	62.012456	10.125696	0.432367
min	1996.000000	495.000000	1.000000	0.000000	20.800000	0.000000
25%	2016.000000	8999.000000	9987.000000	30.000000	52.300000	1.000000
50%	2017.000000	11291.000000	18242.500000	145.000000	58.900000	1.200000
75%	2018.000000	15299.000000	31060.000000	145.000000	65.700000	1.500000
max	2060.000000	54995.000000	177644.000000	580.000000	201.800000	5.000000

[20]
```
df["price"].min()
df["price"].max()
df["price"].mean()
df["price"].median()
```
```
11291.0
```

[21]
```
df["price"].median()
```
```
11291.0
```

[22]
```
df["price"].mean()
```
```
np.float64(12279.534843593454)
```

[23]
```
df["price"].max()
```
```
54995
```

[24]
```
df["price"].min()
```
```
495
```

```
df["mileage"].min()
df["mileage"].max()
df["mileage"].mean()
df["mileage"].median()
```

```
18242.5
```

```
df["mileage"].min()
```

```
1
```

```
df["mileage"].max()
```

```
177644
```

```
df["mileage"].mean()
```

```
np.float64(23362.608760992985)
```

```
df["mileage"].median()
```

```
18242.5
```

```
df["year"].min()
df["year"].max()
df["year"].mean()
df["year"].median()
```

```
2017.0
```

```
df["year"].min()
```

```
1996
```

```
df["year"].max()
```

```
2060
```

```
df["year"].mean()
```

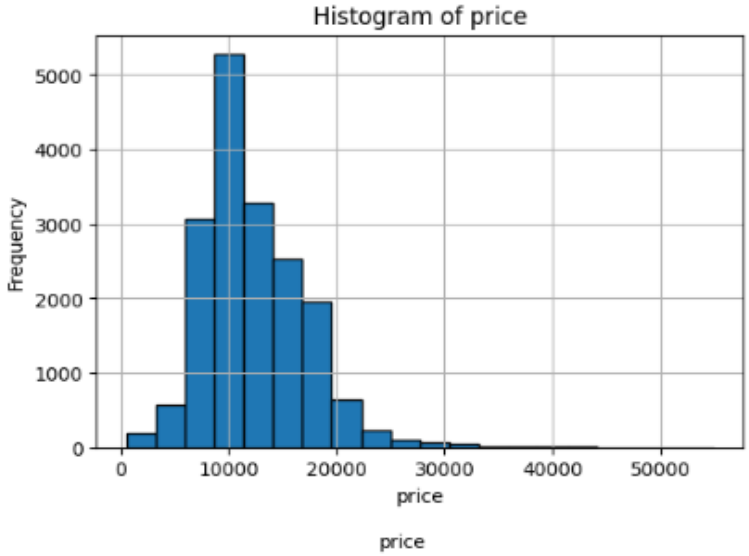
```
np.float64(2016.8664699988867)
```

```
df["year"].median()
```

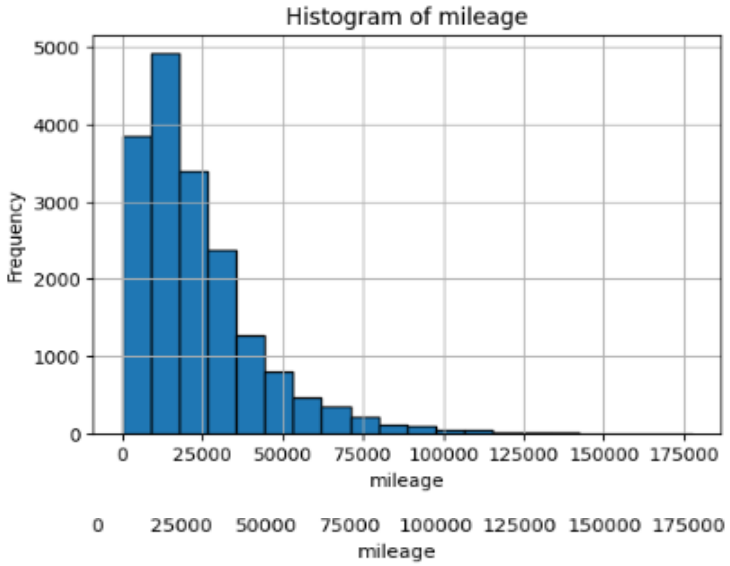
```
2017.0
```

4)

...

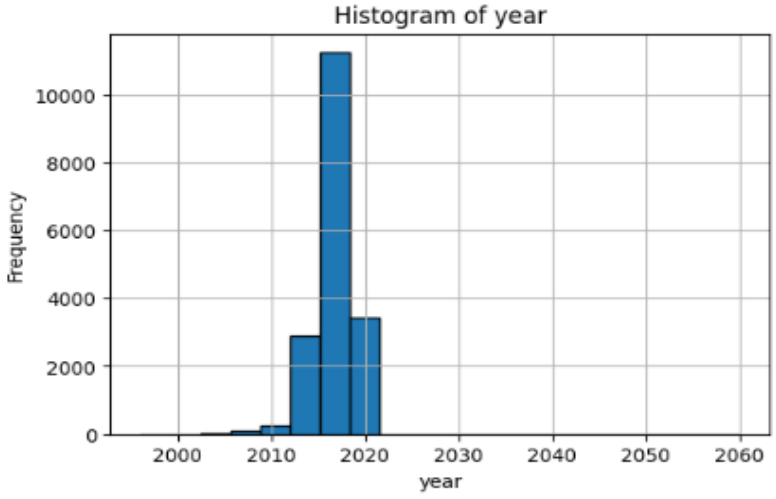


price

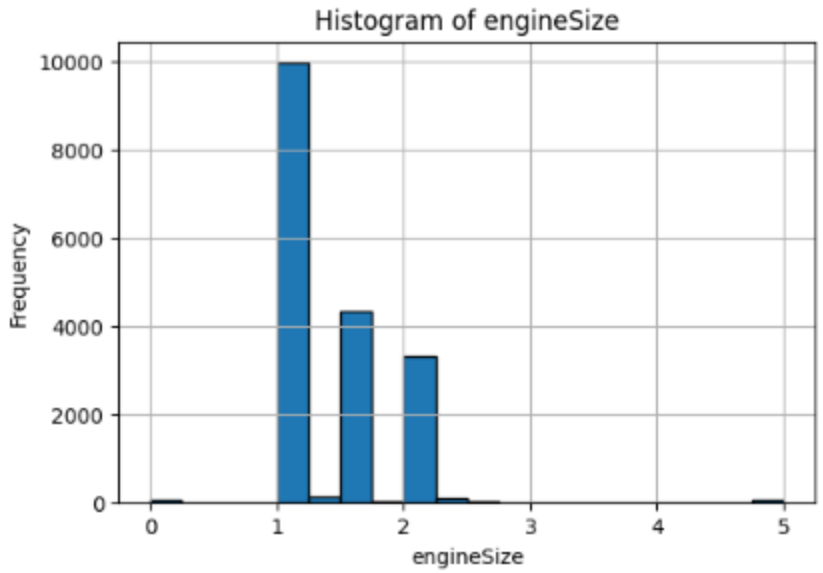


0 25000 50000 75000 100000 125000 150000 175000

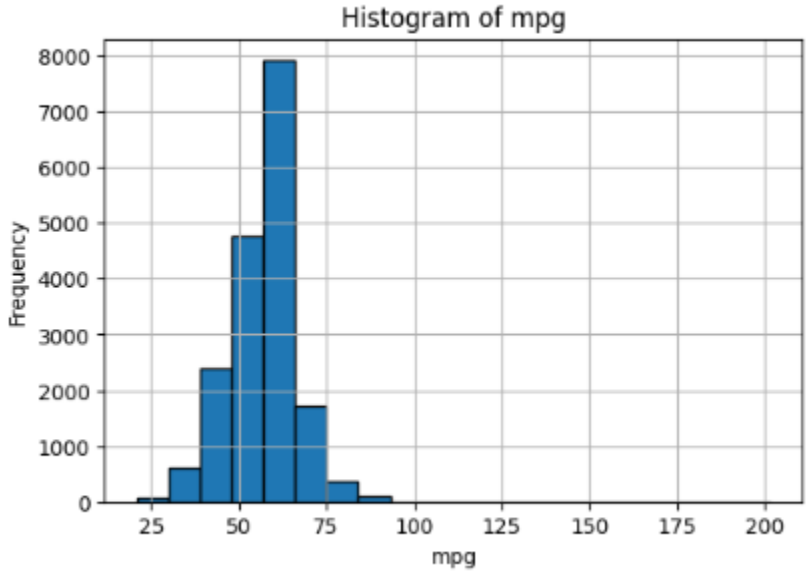
mileage



year


Histogram of engineSize
Frequency
engineSize
0
2000
4000
6000
8000
10000
0
1
2
3
4
5


engineSize


Histogram of mpg
Frequency
mpg
0
1000
2000
3000
4000
5000
6000
7000
8000
25
50
75
100
125
150
175
200

5) Clearly identify and list:

Independent Features (Input Features)

Dependent Feature (Output/Target Feature)

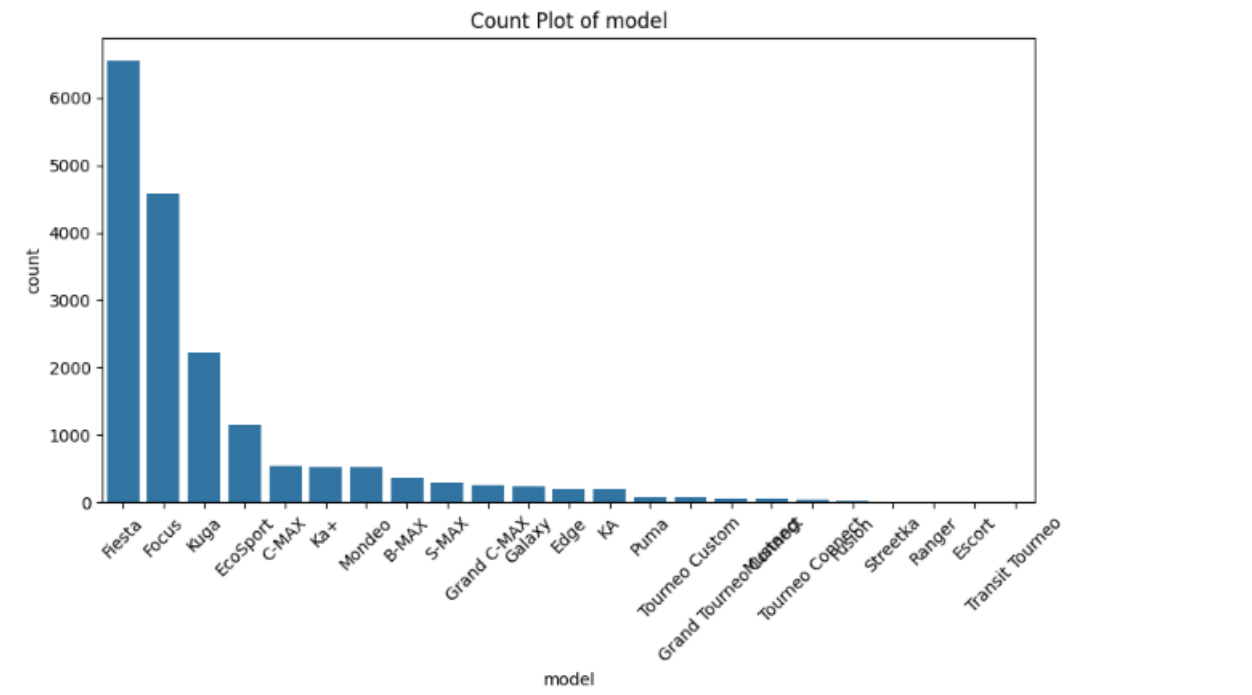
Justify your choice with reasoning.

```
X = df.drop('price', axis=1)
print(X.columns)
```

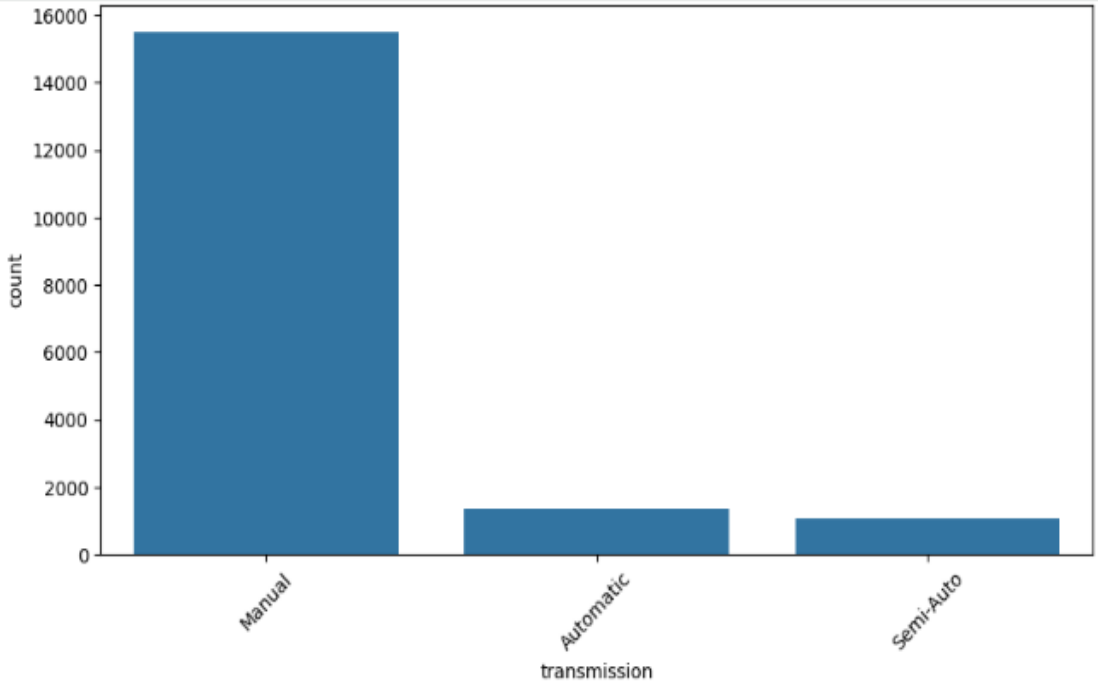
```
Index(['model', 'year', 'transmission', 'mileage', 'fuelType', 'tax', 'mpg',
       'engineSize'],
      dtype='object')
```

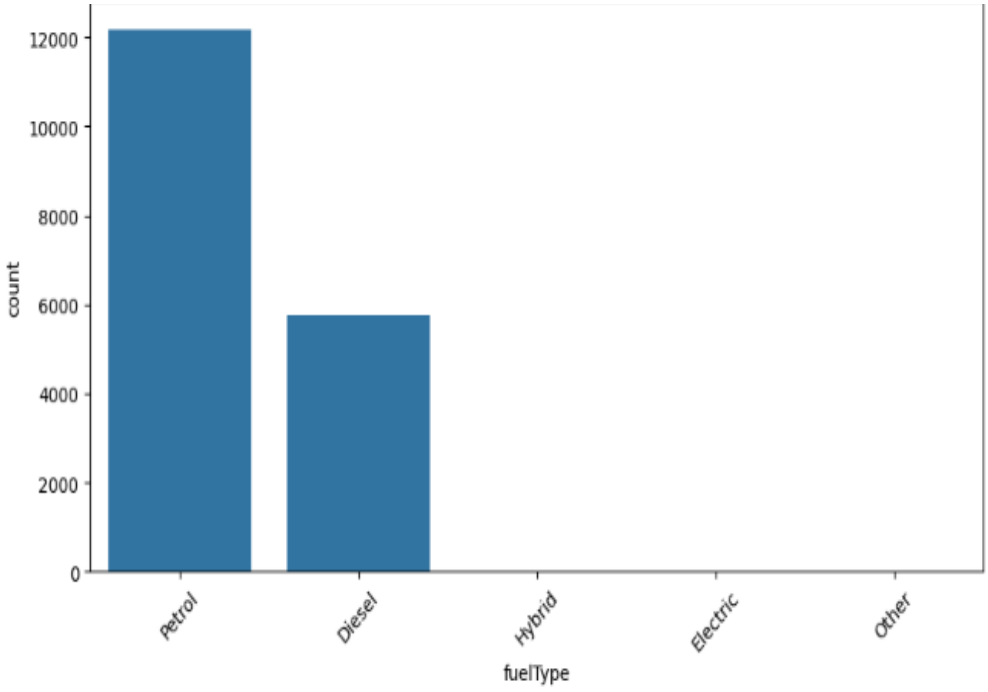
6) Create count plots for all categorical columns (fuelType, transmission, model, etc.) using seaborn.
Analyze which categories are most common.

Write insights about market trends based on these plots in comments.



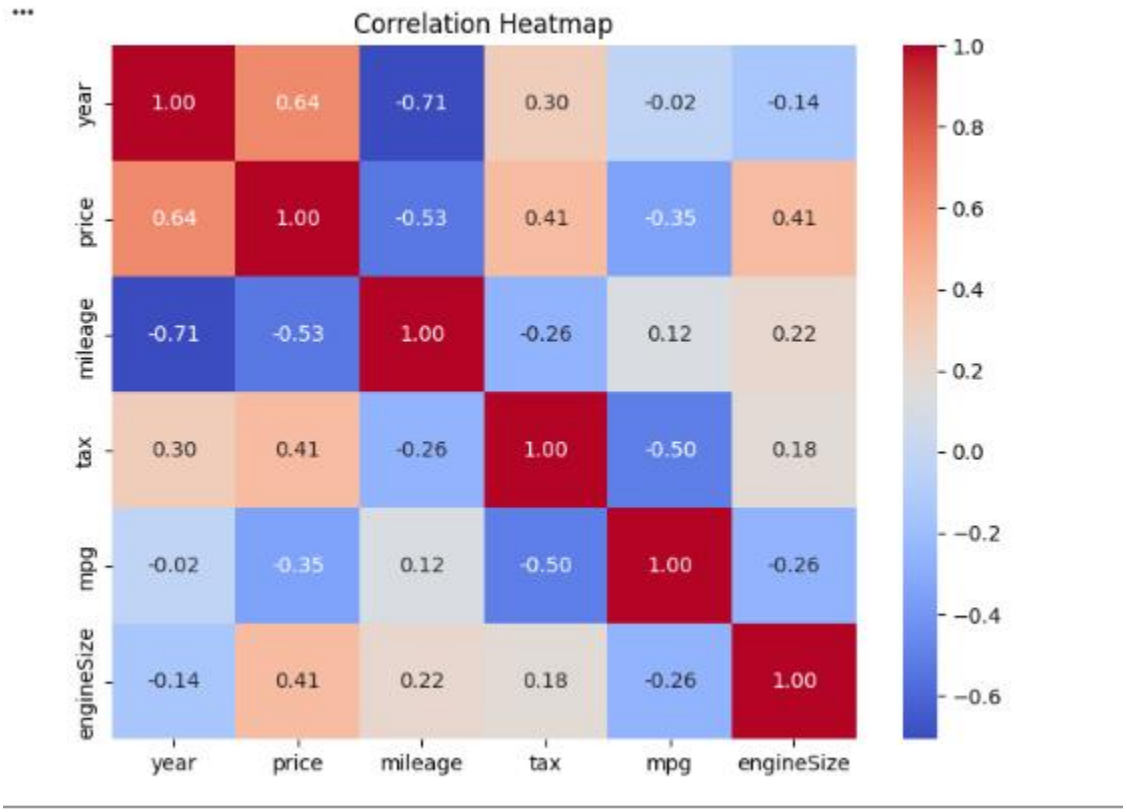
```
plt.show()
```


16000
14000
12000
10000
8000
6000
4000
2000
0
count
Manual
Automatic
Semi-Auto
transmission


12000
10000
8000
6000
4000
2000
0
count
Petrol
Diesel
Hybrid
Electric
Other
fuelType

7)

```
plt.title("Correlation Heatmap")
plt.show()
```



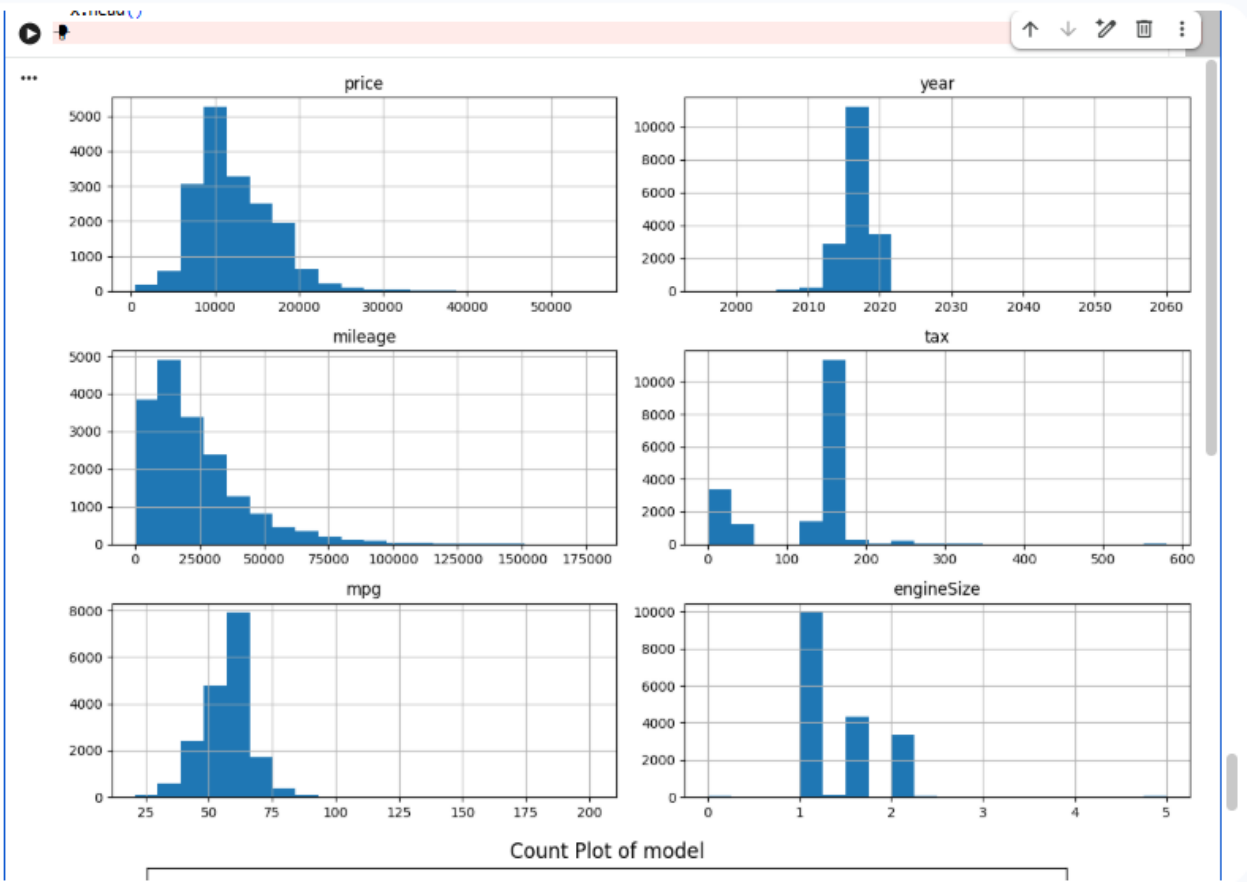
8)

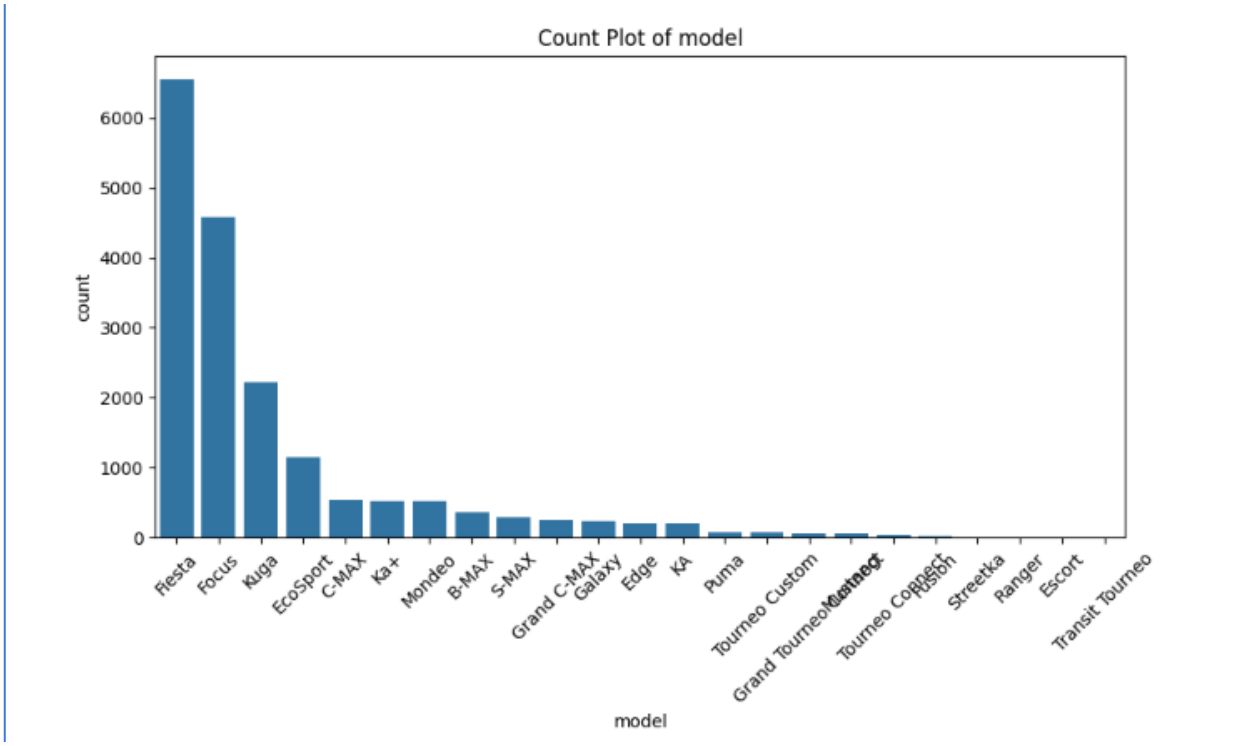
```
from sklearn.preprocessing import StandardScaler
X = df.drop('price', axis=1)
X_numeric = X.select_dtypes(include=['int64', 'float64'])
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_numeric)
X_scaled = pd.DataFrame(X_scaled, columns=X_numeric.columns)
X_scaled.head()
```

	year	mileage	tax	mpg	engineSize
0	0.065128	-0.380998	0.591358	-0.020442	-0.811386
1	0.552866	-0.733359	0.591358	-0.020442	-0.811386
2	0.065128	-0.560132	0.591358	-0.020442	-0.811386
3	1.040605	-0.662640	0.510727	-1.738890	0.345070
4	1.040605	-1.123724	0.510727	-0.909294	-0.811386

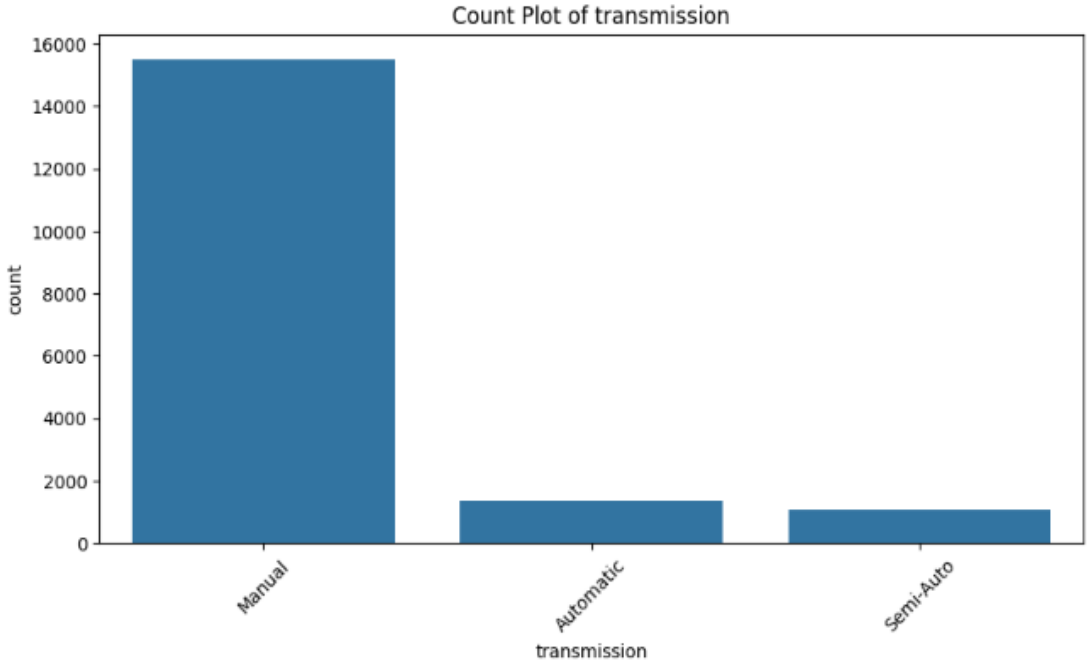
Next steps: Generate code with X_scaled | New interactive sheet

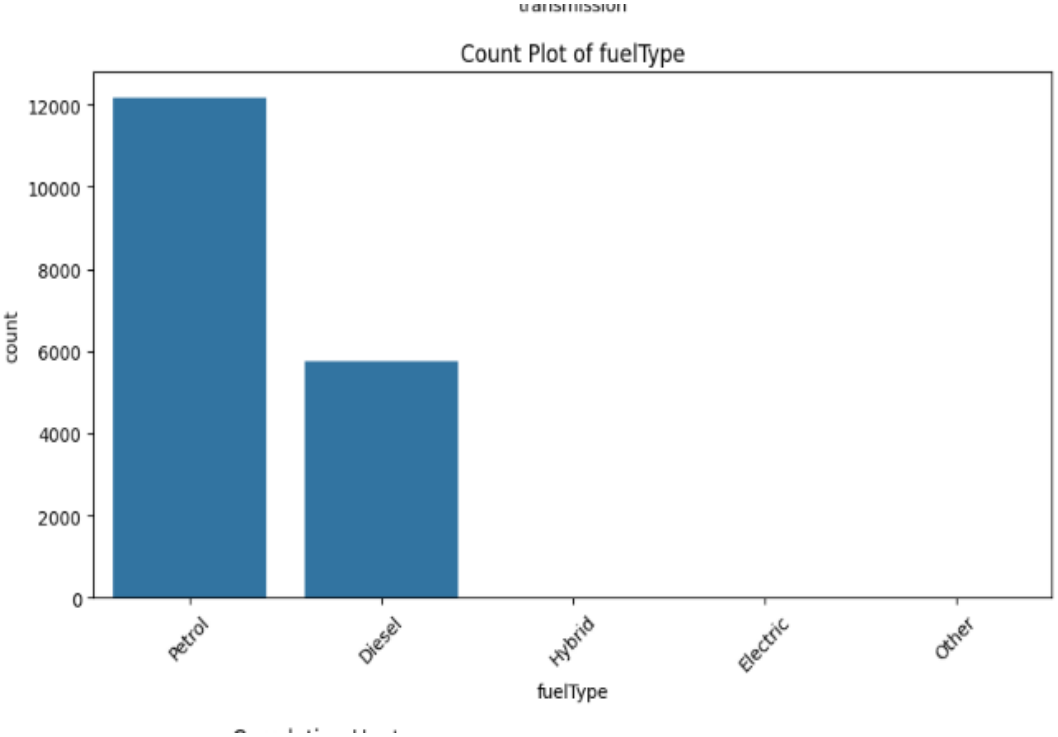
9)

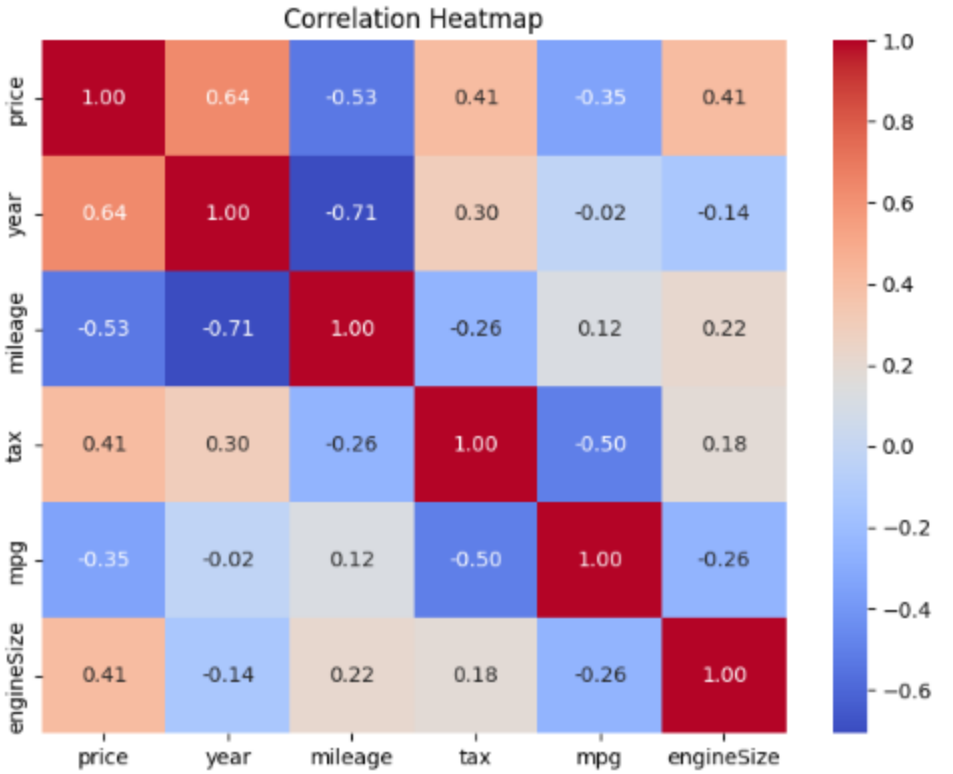

price
year
mileage
tax
mpg
engineSize
Count Plot of model


Count Plot of model
count
model
Fiesta
Focus
Kuga
EcoSport
C-MAX
Ka+
Mondeo
B-MAX
S-MAX
Grand C-MAX
Galaxy
Edge
KA
Puma
Tourneo Custom
Grand Tourneo
Tourneo Connect
Fusion
Streetka
Ranger
Escort
Transit Tourneo

model


Count Plot of transmission
16000
14000
12000
10000
8000
6000
4000
2000
0
count
Manual
Automatic
Semi-Auto
transmission


Count Plot of fuelType
12000
10000
8000
6000
4000
2000
0
count
Petrol
Diesel
Hybrid
Electric
Other
fuelType



	model	year	transmission	mileage	fuelType	tax	mpg	engineSize
0	5	0.065128	0	-0.380998	4	0.591358	-0.020442	-0.811386
1	6	0.552866	1	-0.733359	4	0.591358	-0.020442	-0.811386
2	6	0.065128	1	-0.560132	4	0.591358	-0.020442	-0.811386
3	5	1.040605	1	-0.662640	4	0.510727	-1.738890	0.345070
4	5	1.040605	0	-1.123724	4	0.510727	-0.909294	-0.811386